TBdecompile Notes

ToolBook II 8.1 — 2026-10-12

Consolidated decompile findings: object create path, graphics painters, OpenScript/system books, symbol maps, open questions, and debugger notes.

TBdecompile — Decompile Notes Index

Consolidated notes from the ToolBook II 8.1 reverse-engineering work.

Sources: BAS/RTM decompiles, SoftICE field notes, tbwp0002–0004, and the

NIDA system-book analysis.

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Key confirmed facts

- Object size is 0x76 bytes. Bounds are four words at object+0xF.
- `CDBSENDCREATE` receives bounds as parameters; it does not parse `14 00 00`.
- The bounds writer is `FUN_12c8_0ef8`. The disk reader is a caller of `CDBSENDCREATE`.
- Polygon painter: `FUN_1408_0376`. Scale: `FUN_1008_01dd` (stored/15 at DPI 96).
- OpenScript decoder handles the NIDA navigation patterns. tb80r uses the same bytecode.
- moralrecordings FIDBs produce no CRT renames. Compiler is Microsoft C (R60xx).

Still open

- Disk `14 00 00` → object+0xF expansion rule (caller of `CDBSENDCREATE`).
- Curve control structure (P420).
- Arc disk layout match.
- Picture id → DIB binding.
- RTM book-open path (dies before open in Unicorn).
- CMP W16 segment-not-present fault.

Object create path and bounds

From the honest unpack (`tbwp0001/TB80BAS_noghidra.DLL`) loaded in Ghidra.

Source writeup: `tbwp0002/bas-create-path.md`.

Object layout

• Object size is **0x76** bytes. `CDBNEWOBJECT` indexes them as `param_6 * 0x76`.
• Bounds are four words at **object+0xf, +0x11, +0x13, +0x15** (left, top, right, bottom).
• Flags byte at object+0x1f. Bits set from the source structure at +0x0a, +0x0c, +0x0e.
• Source-structure fields copied by `FUN_12c8_0ef8`:
- source+0x3e / +0x40 / +0x42 / +0x44 → object+0x17 / +0x19 / +0x1b / +0x1d
- source+0x06 → object byte at offset given by a type table
- source+0x04 → pattern lock

Create path

```
CDBSENDCREATE          (12c8:00bc, Ordinal_51)
  ■■ FUN_12c8_0850     (12c8:0850)   one caller only
       ■■ FUN_12c8_0e32 (12c8:0e32)  allocator
            ■■ FUN_12c8_0ef8 (12c8:0ef8)  field copier / bounds writer
```

`FUN_12c8_0850` is not the record reader. It locks context, calls the allocator, and tears the object down on error. `ENTER 0x4,0x0`.

CDBSENDCREATE type dispatch

`param_5` is the type code. The string arithmetic in the decompile is Ghidra matching constants against `s_CTbkSimpleOLEDispatch_CallMethod`.

param_5	path
0x1014, 0x1015	`FUN_12c8_0850` — new object
0x1021, 0x1022	`FUN_12c8_0a86`
0x3018–0x301b	bounds-checked with `CDBCHECKBOUNDDTL`, then `FUN_12c8_0e08`
0x3022	`FUN_12c8_0d40`
else	`Ordinal_115`

Parameters: `param_3`/`param_4` are the bounds pair (clamped against 32000 and 0x8300). `param_6`/`param_7` are compared to the current object id. `param_9` is an error out-byte.

FUN_12c8_0ef8 bounds write

Two paths write the four bounds words:

1. When `param_2` is null, copy from globals set from `param_4`/`param_5`:

```
object+0x0f = DAT_1630_01fe;  // param_4
object+0x11 = DAT_1630_0200;  // param_5
object+0x13 = DAT_1630_0202;  // param_4
object+0x15 = DAT_1630_0204;  // param_5
```

2. In the `switch(param_6)` default case, copy four words from

`param_2` into the same offsets.

`param_6` is the object type:

param_6	path
0x0f, 0x10, 0x13, 0x14	polygon-like; `OBJNEWNEAR(param_1 << 2)` for a point array
0x11, 0x12	break (arc/pie; 0x11 matched the SoftICE arc)
0x24	special; also copies four words from `param_2`
default	copies `param_2[0..3]` into object+0xf..+0x15

Property table

`FUN_1058_0a98` at **1058:0a98**. Builds the property-id list.

First write of `0x4004` is at decompile line 30313.

Nine writes of `0x4004` in the function.

Ids written: `0x4004` (bounds), `0x4009`, `0x400d`, `0x4015` (idNumber), `0x401a`, `0x401b`, `0x401c`, `0x4020`–`0x4022`, `0x4023` (script), `0x4027`, `0x402b`, `0x4031`–`0x4033`, `0x4034` (vertices), `0x4035`.

Still open

`CDBSENDCREATE` receives bounds as parameters. It does not parse a `14 00 00` record. The record-to-bounds reader is a caller of `CDBSENDCREATE`, either in BAS or in the book-open path in RTM/Tbload.

Next step: list XREFs on `CDBSENDCREATE`. The caller that passes a pointer into a page blob is the reader.

Graphics painters (TB80BAS)

Confirmed from the decompile and SoftICE traces.

Painters

Function	Segment:offset	Role	Notes
`FUN_1408_0376`	1408:0376	PaintPolygon	Count at object+0x30. Calls `SetupPaintRect`, `ScaleTwipsToScreen` on DS:30AC, then `GDI!POLYGON`.
`FUN_1410_0066`	1410:0066	PaintRectangle	Reads bounds at object+0xF, calls `GDI!RECTANGLE`.
`FUN_1008_01dd`	1008:01dd	ScaleTwipsToScreen	`screen = (stored * DPI + 720) / 1440`. At DPI 96: stored/15.
`FUN_10a0_0558` / `FUN_10a0_0458`	10a0	Arc helpers	Live object type 0x11, four twip pairs, fill at +0x18.
`FUN_1408_00d8`	1408:00d8	SetPointArray	Runtime property 0x4050 setter. Not on disk.

On-disk polygon (solved)

• Signature: `14 00 00` subtype-0, `+0x0c = 04 00`.
• Fill dword at +4.
• Point count at +0x1A.
• x,y twip pairs start at +0x22.
• Screen = twips/15.
• 96 such records on 13 pages in 9e954d01; P430 has 22.

Still open

Item	Status
Curve control structure (P420)	Points exist at page+0xA13. No count/header recovered. Render is a tangle.
Arc disk layout	Live object known. No matching `14 00 00` pattern tied on disk.
Picture id → DIB	641 BMPs carved. Widget field that holds the id not identified. TB80BMP ordinal 3 is the reader.
Computed shapes (brackets)	Segment 0x5B builds them from bounds. Not stored points.

Scale function (recovered)

```
// FUN_1008_01dd — simplified
// For each x,y pair:
//   screen_x = (stored_x * DAT_1630_01a6 + 0x2d0) / 0x5a0
//   screen_y = (stored_y * DAT_1630_01a8 + 0x2d0) / 0x5a0
// DAT_1630_01a6 / 01a8 = 96 on the observed boot → /15.
```

OpenScript and NIDA system books

OpenScript decoder

• Keyword table is resource 0x77 (type 100, id 119). Offset table at byte 2 + count*4.
• High-bit opcodes (`0xE0`, `0xF0`, `0xFF`, …) emit whitespace, CR/LF, or repeats.
• Back-references (`0xC0`–`0xFF`) copy from the compressed input stream.
• Decoder stops at the first standalone `end`.
• Handles `go to page`, `send next`/`previous`, basic conditionals, and system properties.
• Corpus: 76 handlers from 9e954d01.tbk, 100 buttonClick handlers from nida81.tbk.

tb80r.sbk compression

• Uses the same bytecode as the lesson books.
• Difference: the handler name prefix ("to handle" / "to get") is also compressed.
• 33 handlers extracted (`tbwp0004/tb80r_handlers.json`).
• Encoding rules are identical; byte distribution differs (more 0xFF/0xE0).

NIDA system books

File	Role
`nida81.tbk`	Shared shell. 43 objects, pages P20000–P22130. 100 buttonClick handlers.
`nc81base.sbk`	Resource container. Object-type table. No OpenScript.
`nc81recs.sbk`	Resource container. No OpenScript.
`nc81test.sbk`	Resource container. No OpenScript.
`nc81trnr.sbk`	Resource container. No OpenScript.
`nc81int.dll`	47 exports. Trainer/comms (IntCheckComm*, voltages, switches, faults).
`nc81rec.dll`	Session recording (RecStartSession, RecWriteResponse, RecBookmarkSession, …).

Lessons call `initializebase`, `initializerecs`, `initializetest`, `openSession`, `gotoPageList`.

These resolve to nida81.tbk handlers or the two DLL exports. The `.sbk` files hold no script.

Still open

- Full call inventory across the ATT Master_3 set (not just one lesson).
- Minimal stubs for the closed NIDA call set.
- Picture-id binding and remaining draw-object disk layouts.

Symbol maps

Generated from the Ghidra decompiles. Source: `tbwp0003/`.

File	Entries	Source
`TB80BAS.map`	3799	TB80BAS_noghidra.DLL.c
`TB80RTM.map`	2669	TB80RTM.EXE.c

Format

```
SSSS:OOOO name
```

`SSSS` is the NE segment index from the decompile (for example `13D0`).

`OOOO` is the offset inside that segment. This is not a live selector.

The selector changes every boot. Match the segment index to the live selector in the debugger, then use the offset.

Named exports (`CDBSENDCREATE`, `Ordinal_51`, etc.) replace the `FUN_` name when the decompile attached one. Everything else keeps the Ghidra name.

SoftICE

SoftICE does not load this MAP directly.

1. Break on the BAS ordinal and walk back (no symbols needed):

```
EXP TB80BAS
BPX TB80BAS.Ordinal_51
```

2. Or convert the MAP to a SYM if you have a converter.

Do not use `BAS(82)` or similar. That is not a SoftICE address.

Use `MAP TB80BAS`, read the selector for the segment, then `BPX :offset`.

WDW

WDW does not load this MAP as Watcom debug info. Use it as a lookup:

1. Note the live selector for the NE segment.
2. Find the function in the MAP.
3. Set the breakpoint at `selector:offset`.

Or break on BAS ordinal 51 and read the return address.

Limits

- Offsets are from the unpacked image. They match the decompile.
- No types, no locals, no line numbers. Names and addresses only.
- The segment index must be matched to the live selector by hand.

Open questions

Bounds

- Disk `14 00 00` record → object+0xF expansion rule.
- `CDBSENDCREATE` receives bounds as parameters. The reader is a caller.
- Next: XREFs on `CDBSENDCREATE` in BAS or the RTM/Tbload open path.

Graphics

- Curve control structure (P420 flame). Points exist; order/structure wrong.
- Arc disk layout. Live object (type 0x11) known; disk match missing.
- Picture id → DIB binding. BMPs carved; widget field unknown.
- Computed shapes (brackets, lines) once bounds exist.

OpenScript / system

- Full NIDA call inventory (initialize*, session, gotoPageList).
- Stubs for the closed set that appears in the corpus.
- Modern CMP expander (standalone, no W16). Test corpus already extracted.

Engine

• RTM book-open path. Unicorn hosts die before a book open (FFFF:0158 / unbound far call).
• CMP W16 segment-not-present fault (001c:00000015). Needs live selector capture.
• TBKX export via ToolBook 11.5. Prospective; no samples in tree yet.

Closed

• FIDB trial. moralrecordings DOS/Win16 databases produce zero CRT renames.
• Compiler is Microsoft C (R60xx strings). VC++ 1.52c / MSC 7.0 already covered.
• SoftICE `BAS(nn)` syntax does not work. Use MAP → selector.

Debugger notes (SoftICE / WDW)

From field notes and the corrected breakpoint discussion.

SoftICE standing orders

• `BC *` before the next breakpoint. A live `GDI!POLYGON` breakpoint freezes SoftICE if the page scrolls.
• Selectors change every boot. Copy offsets. Remap selectors.
• `MAP TB80BAS` (or RTM, BMP, …) to get the live selector for a segment.
• Segment numbers in MAP output are hex.
• Do not search all of memory. Wide 80 65 searches can GPF the VM.
• A dump that is the SoftICE window or the DOS device chain (CON/AUX/PRN) missed.

Correct breakpoint form

```
BC *
MAP TB80BAS
; read the selector for the segment you want (e.g. 0x82)
BPX <selector>:0376 DO "D SS:BP-4 L 4; D DS:30AC L 40"
G
```

Invalid: `BPX BAS(82):0376`

SoftICE does not accept that notation.

Useful offsets (stable)

Function	Segment:offset	Notes
PaintPolygon	1408:0376	Object at [BP+04], count at +0x30, points at +0x32
PaintRectangle	1410:0066	Bounds at object+0xF
ScaleTwipsToScreen	1008:01dd	
CDBSENDCREATE	12c8:00bc	Ordinal_51
FUN_12c8_0850	12c8:0850	Allocator caller
FUN_12c8_0ef8	12c8:0ef8	Bounds writer

WDW

• Use the `.map` files as a lookup. Note the live selector, then set `selector:offset`.
• Or break on BAS ordinal 51 and read the return address to land in the caller.

Dump cautions

• Real dumps: page.bin01511 (ASYM_TpID), p400.bin0157 (JBO signature), dump-001/002 (Viewer/System).
• Junk: console logs, DOS device chain, name-slot only.
• ICEDUMP 6.021 needs kernel32 and user32 exports in winice.dat. Memory-dump recipe is not repeatable.

BAS export → decompile mapping

Export	Decompile line	Map address
`ANSILSTRNCMPI`	—	
`ASYM_LOCALANDHELPCBK`	—	
`CDBBOOKOFVALUE`	—	
`CDBCHECKBOUNDDTL`	—	
`CDBCHUNKEND`	—	
`CDBCHUNKFIND`	138575	
`CDBCHUNKLAST`	—	
`CDBCLASSTABLEGETBEHAVIOR`	125154	
`CDBCLEARBOOKREFS`	—	
`CDBCLIENTSTATUS`	—	
`CDBCLONEINSTANCE`	25024	1038:08FC
`CDBCOPYOBJECTVALUE`	41627	1088:27E0
`CDBCREATEBKGNDREF`	—	
`CDBCREATEBOOKREF`	28347	1048:0DDA
`CDBCREATEGRAPHICREF`	37711	1070:0000
`CDBCREATEINSTANCE`	24974	
`CDBCREATEPAGEREF`	25913	1010:211C
`CDBCREATETHREAD`	—	
`CDBDEREFVALUE`	—	
`CDBDESTROYINSTANCE`	—	
`CDBDESTROYTHREAD`	—	
`CDBDISPLAYSTATE`	12804	1008:4B4A
`CDBEDITBOOK`	—	
`CDBEXCLUSIVELOCK`	—	
`CDBFILECLONE`	—	
`CDBFILECLOSE`	—	
`CDBFILEMERGE`	—	
`CDBFILENEW`	27532	
`CDBFILEOPEN`	27592	1028:1226
`CDBFILEPATH`	28002	1048:0B46
`CDBFILESAVE`	27675	
`CDBFILESAVEAS`	27751	
`CDBFILESTATUS`	—	

Export	Decompile line	Map address
`CDBFORMATPLERR`	—	
`CDBGETEVENTLIST`	124383	1278:1A72
`CDBGETFOCUSPROP`	32396	
`CDBGETHANDLERBREAKCOUNT`	52502	1028:203C
`CDBGETHANDLERBREAKS`	52551	
`CDBGETSELECTPROP`	33826	1008:0A76
`CDBGETVALUE`	38475	
`CDBGETVALUEEX`	38488	
`CDBIMAGESTORE`	—	
`CDBINIT`	—	
`CDBINTERNALTESTCOMMAND`	—	
`CDBLAYEROFVALUE`	—	
`CDBLOCKFONT`	130353	
`CDBLOCKVALUE`	—	
`CDBMETRICS`	—	
`CDBMULTISAVE`	28272	
`CDBNEWOBJECT`	131992	12C8:0850
`CDBPAINTCLIENT`	13056	
`CDBPARENTVALUE`	—	
`CDBQUERYFILEPATH`	38322	1078:036C
`CDBQUERYMULTISAVE`	28108	1048:0B6A
`CDBQUERYVALUE`	—	
`CDBREFVALUE`	—	
`CDBREGISTERINSTWINDOW`	25296	
`CDBREGISTEROLE2WINDOW`	142287	1310:0292
`CDBREGISTERVBXFORM`	—	
`CDBRESETPLERR`	—	
`CDBSAMEVALUEP`	37953	1040:06D6
`CDBSCRIPTSEARCH`	—	
`CDBSEARCHINIT`	—	
`CDBSENDCREATE`	131896	
`CDBSENDEDITLEAVETOALLFILEVIEWERS`	—	
`CDBSENDEDITLEAVETOALLVIEWERS`	—	
`CDBSENDFOCUS`	32291	1058:2ACA
`CDBSENDSELECT`	33879	
`CDBSENDUNDO`	—	
`CDBSENDVALUE`	38701	
`CDBSENDVALUEEX`	38712	
`CDBSETFOCUS`	32439	1060:143E
`CDBSETFOCUSPROP`	32414	1060:11E8
`CDBSETIMAGEMEMORYMODEL`	—	
`CDBSETINTERNALTESTSTATE`	—	
`CDBSETPLERR`	81300	
`CDBSETSELECTPROP`	34107	1030:00B0
`CDBSETVALUE`	38889	
`CDBSETVALUEEX`	38900	1048:2F0E

Export	Decompile line	Map address
`CDBSYNCSCREENS`	—	
`CDBTHREADFROMINSTANCE`	—	
`CDBUNIONDAMAGE`	—	
`CDBUNREGISTERVBXFORM`	—	
`CDBVALUEBOOKFROMSTRING`	90045	
`CDBVALUEBOOKTOSTRING`	90104	
`CDBVALUEGRAPHICFROMSTRING`	90406	
`CDBVALUEGRAPHICTOSTRING`	90549	
`CDBVALUELAYERFROMSTRBOOK`	90173	
`CDBVALUELAYERFROMSTRING`	90158	
`CDBVALUELAYERTOSTRING`	90393	1028:0466
`CDBVALUENEWBOOKREF`	—	
`CDBVALUENEWGRAPHICREF`	89968	
`CDBVALUENEWLAYERREF`	89850	
`CDBVALUERESOURCEFROMSTRING`	90315	
`CDBVIEWBKGND`	—	
`CDBVIEWBOOK`	—	
`CDBVIEWNONE`	—	
`CDBVIEWPAGE`	25726	
`CDBVIEWPAGEVALUE`	25936	
`CDROMOBJINFO_ADDPROC`	148404	1350:370E
`CDROM_RECORDERROR`	145779	
`CDROM_SELFTEST`	—	
`CLASSTABLEGETCLASSEXTRABYTES`	124768	
`CLASSTABLEGETCLASSEXTRABYTESSIZE`	124742	1280:0AC8
`CLASSTABLEGETCLASSID`	124530	1280:2C66
`CLASSTABLEGETCLASSIDFROMMANGLEDNAME`	124585	1278:19B4
`CLASSTABLEGETCLASSLIBNAME`	124692	
`CLASSTABLEGETOBJECTSIZE`	124729	
`COLORHLSTORGB`	155991	
`COLORSTRINGS`	155560	1370:0256
`COMBOLBOXWNDPROC`	108089	
`COMBOWNDPROC`	—	
`COMPAREASNAME`	—	
`CONVERTTORUNTIMEMEDIAPATH`	114241	
`COPYDITHEREDPALETTE_8BIT`	—	1170:4098
`CREATEDIBMASK`	—	
`CREATEDITHEREDHEADER_8BIT`	—	
`DRAWDDSCURVESEG`	—	
`GETBITMAPINFO`	—	
`GETCDBID`	—	
`GETCLASSDATA`	—	
`GETDATA`	—	
`GETDIBCOLORCOUNT`	—	
`GETDIBSIZE`	—	

Export	Decompile line	Map address
`GETHDC`	—	
`HCACHEPURGE`	—	
`HDLRINFO_ISVALID`	—	
`HEAPADDTOCLEANUPLIST`	—	
`HEAPCREATEFRAME`	—	
`HEAPDEREGISTER`	—	
`HEAPENTERCRITSECT`	—	
`HEAPGETBPFRAME`	—	
`HEAPGETHANDLER`	—	
`HEAPLEAVECRITSECT`	—	
`HEAPMARKFRAME`	—	
`HEAPMARKPOINTER`	—	
`HEAPPOINTERTOVALUE`	—	
`HEAPREGISTER`	—	
`HEAPRELEASEFRAME`	—	
`HEAPREMOVEFROMCLEANUPLIST`	—	
`HEAPSWEEPFRAME`	—	
`HEAPVALUETOFRAME`	94020	
`HEAPVALUETOPOINTER`	—	
`HUGEGETDATA`	74213	1168:118A
`IMBITBLT`	—	
`IMBLACKOUTBMP`	—	
`IMDRAW`	—	
`IMDRAWLEVELS`	13353	1020:16DE
`IMINVERTBMP`	—	
`IMSHOWBMP`	—	
`LIONELDEREFBOOK`	—	
`LIONELREFBOOK`	—	
`LOCALANDHELPHOOK`	20418	
`LOCKPATTERN`	43779	
`LOOKUPSTDPROPBYNAME`	—	
`LPSTRTODOUBLE`	86527	11C8:058E
`MATHERRFLOAT`	—	
`MATHERROR`	—	
`OBJARENACHECK`	72286	
`OBJFILECLOSE`	—	
`OBJFILECOPY`	75987	
`OBJFILEOPEN`	76162	11A8:04F6
`OBJFREE`	75619	
`OBJFREENEAR`	75662	
`OBJFREENEAREX`	75710	1160:0774
`OBJFREESEG`	—	
`OBJGETFILE`	—	
`OBJGETHDRLONG`	—	
`OBJINIT`	—	
`OBJLOCK`	75082	
`OBJLOCKNEAR`	75136	

Export	Decompile line	Map address
`OBJLOCKSEG`	—	
`OBJNEWNEAR`	75226	1178:114C
`OBJNEWNEAREX`	75270	
`OBJNEWSEG`	75325	
`OBJOOTOOH`	74975	1178:151E
`OBJOPTOOH`	—	
`OBJPRINCIPALOO`	—	
`OBJPRINCIPALOP`	—	
`OBJREALLOCNEAR`	75365	
`OBJREALLOCNEAREX`	75413	
`OBJTOUCH`	—	
`OBJUNLOCK`	—	
`PALENTRYTORGBQUAD`	—	
`PALENTRYTORGBTRIPLE`	—	
`PATTERNDITHER_8BIT`	—	1598:001A
`PROFILE_SETLOGFILE`	—	
`PSTRTODATEWITHFMT`	152434	1358:13B4
`PSTRTOTIMEWITHFMT`	152259	1358:126E
`RESFONTINSTALL`	—	
`RESMGRCHOOSEFILE`	—	
`RESMGRCLOSEBOOK`	—	
`RESMGRCONNECT`	—	
`RESMGRCOPY`	—	
`RESMGRDELETE`	—	
`RESMGRDISCONNECT`	—	
`RESMGRENUM`	22381	
`RESMGREXPORT`	20962	10D0:1F5C
`RESMGREXPORTBYHANDLE`	21076	
`RESMGREXTRACTGDI`	21523	1028:23EC
`RESMGRGETNAME`	22096	
`RESMGRGETREFCNT`	22579	1008:0422
`RESMGRIMPORT`	20699	
`RESMGROPENBOOK`	20177	1028:2770
`RESMGRQUERYINFO`	—	
`RESMGRREPLACE`	—	
`RESMGRSENDMESSAGE`	—	
`RESMGRSENDMSG`	—	
`RESMGRSETNAME`	22107	1028:27D2
`RGBQUADTOLOGPALETTE`	—	
`SAMPLETIMER`	—	
`SETUPLAYERSTATE`	—	
`STACKELEMENT`	154344	
`STACKFREE`	—	
`STACKGETSIZE`	—	
`STACKPOP`	154759	
`STACKPUSH`	154537	
`STACKPUSHONE`	154643	

Export	Decompile line	Map address
`STACKSETELEMENT`	154487	
`STACKSLIP`	154806	1368:1586
`STARTTIMER`	—	
`STOPTIMER`	—	
`STRINGFROMDATEWITHFMT`	152079	
`STRINGFROMTIMEWITHFMT`	151951	
`TBHEAPALLOC`	93620	
`TBHEAPCLEARCRITSECT`	—	
`TBHEAPFREE`	—	
`TBHEAPRESTORECRITSECT`	—	
`TBHEAPSAVECRITSECT`	—	
`TBKVERSION`	—	
`TBX32MSGTHUNK`	—	
`TBXAUTHORMODE`	—	
`TBXAUTOEXECSCRIPTVARIANT`	115293	12F8:0018
`TBXBOOKGETFDESC`	—	
`TBXBOOKGETFINDEX`	—	
`TBXBOOKLOCKFDESC`	115706	
`TBXBOOKUNLOCKFDESC`	—	
`TBXCTOVALUE`	115614	
`TBXCTOVALUEEX`	115625	
`TBXDAMAGEDOBJECT`	115789	12B8:1536
`TBXEXECSCRIPT`	115148	
`TBXEXECSCRIPTDYNAMIC`	115215	
`TBXEXECSCRIPTVARIANT`	115231	1268:0516
`TBXEXECSCRIPTVARIANTSPCL`	115358	1008:09F8
`TBXEXECSCRIPT_CALLBACK`	115128	1268:0488
`TBXFIXPENRECT`	114838	1198:0146
`TBXFONTINFO`	—	
`TBXGET`	128412	1288:070C
`TBXGETBKGNDRGBFILLCOLOR`	115489	
`TBXGETBKGNDRGBFILLCOLOR_CALLBACK`	115477	1398:125C
`TBXGETCLASSID`	114848	1008:01DD
`TBXGETFONTFACEINFO`	115739	1110:0764
`TBXGETPOINTER`	—	
`TBXGETREADERRIGHTCLICK`	—	
`TBXHUGEALLOC`	114915	15E8:021E
`TBXHUGEFREE`	114936	1168:04B4
`TBXHUGEGETDATA`	115015	
`TBXHUGEGETDATA_CALLBACK`	114997	1168:0C7A
`TBXHUGESETDATA`	114975	
`TBXHUGESETDATA_CALLBACK`	114957	1168:0740
`TBXNEARALLOC`	114865	
`TBXNEARFREE`	—	
`TBXNEARREALLOC`	114883	15E8:01C2
`TBXSEND`	128378	1278:1FE2

Export	Decompile line	Map address
`TBXSENDINST`	115536	
`TBXSENDINSTFROMENVP`	115550	
`TBXSENDPOINTER`	—	
`TBXSET`	128460	
`TBXSETFOCUS`	115773	
`TBXSETPOINTER`	—	
`TBXSTREAMOUT`	115826	1008:0417
`TBXVALUECOPY`	115603	
`TBXVALUESTRINGLENGTH`	115696	
`TBXVALUETOC`	115581	
`TBXVALUETOCEX`	115592	
`TBXVERIFYID`	—	
`VALUEARRAYCHKDIMS`	87865	
`VALUEARRAYGET`	87982	
`VALUEARRAYGETVALUE`	88046	
`VALUEARRAYINFO`	87815	
`VALUEARRAYSET`	88133	
`VALUEAS`	84658	
`VALUEASC`	84734	
`VALUEASCEX`	84760	
`VALUEASEX`	84684	
`VALUEATOL`	—	
`VALUECHECK`	92136	
`VALUECOMPAREDATE`	156391	1370:1312
`VALUECOMPARETIME`	156432	
`VALUECONVERT`	84809	
`VALUECONVERTEX`	84835	
`VALUECONVERTTOC`	84985	
`VALUECONVERTTOCEX`	85011	
`VALUEDISPOSE`	—	
`VALUEFREE`	—	
`VALUEFROMSTRING`	—	
`VALUEFROMSTRINGEX`	—	
`VALUEINIT`	—	
`VALUEMARK`	—	
`VALUENEW`	—	
`VALUENEWARRAY`	87648	
`VALUENEWBOOK`	90561	
`VALUENEWCOLOR`	155412	1368:1538
`VALUENEWCOLORTABLE`	155590	11C8:0E64
`VALUENEWCONST`	—	
`VALUENEWCPOINT`	—	
`VALUENEWDATE`	151679	
`VALUENEWDOUBLE`	86392	
`VALUENEWFROMDOUBLE`	86591	
`VALUENEWFROMLONG`	86648	
`VALUENEWGRAPHIC`	91027	11E0:1D9E

Export	Decompile line	Map address
`VALUENEWINTEGER`	—	
`VALUENEWKEYWORD`	96022	1200:016C
`VALUENEWLAYER`	90920	
`VALUENEWLAYEREX`	90904	
`VALUENEWLITDBL`	—	
`VALUENEWLITSTR`	—	
`VALUENEWLONG`	86271	11C8:012E
`VALUENEWPOINT`	—	
`VALUENEWRECT`	153433	1360:0298
`VALUENEWRESOURCE`	90660	
`VALUENEWSTACK`	—	
`VALUENEWSTRING`	87385	
`VALUENEWSTRINGEX`	87452	
`VALUENEWTIME`	151828	
`VALUETOC`	—	
`VALUETOCEX`	—	
`VALUETOSTRING`	—	
`VALUEVALIDATE`	—	
`VBXDESTROYBOOK`	—	
`VBXDESTROYVIEW`	—	
`VBXINIT`	129642	
`VBXNEWBOOK`	—	
`VBXNEWVIEW`	—	
`WEP`	—	
`WEP`	—	
`XCOPYNEAROBJECT`	—	

Matched in decompile: 174 / 337

Mangled C++ exports (OLE/COM, `?NAME@CLASS@@...`) are not listed.
They correspond to the CTbkSimpleOLEDispatch / CDROPTARGET family
and are not resolved to FUN_ names in the current decompile.

Ordinal assignments (from Ghidra listing)

Export	Ordinal	Segment	Offset	Notes
`CDBCHUNKEND`	53	95 (12f0)	0000	Called by `CDBCHUNKFIND`, `CDBCHUNKLAST`. DBCS-aware chunk scanner.
`CDBCHUNKLAST`	56	95 (12f0)	0334	Calls `CDBCHUNKEND` and `CDBSKIPSEP`.
`CDBCHECKBOUNDDTL`	355	90 (12c8)	0000	Bounds range check. Called from `CDBSENDCREATE`, `CDBNEWOBJECT`. Uses device caps (LOGPIXELSX/Y) cached in `DAT_1630_016e` / `DAT_1630_0170`.

CDBCHECKBOUNDDTL detail

• Caches horizontal/vertical twips-per-pixel from `GetDeviceCaps` (indices 0x58 / 0x5a).

• `param_1 == 'X'` checks against horizontal limit; `'Y'` against vertical.

• Returns 1 if the value is within range, 0 otherwise.

• Called at four sites inside `CDBSENDCREATE` and two inside `CDBNEWOBJECT`.

• This is the clamp, not the disk-to-object bounds reader.

`CDBSKIPSEP`	55	95 (12f0)	04b4	Separator skipper. Called by `CDBCHUNKFIND`, `CDBCHUNKLAST`, `CDBCHUNKREMOVE`. Modes: 1 = nonempty check, 2 = CRLF pair, 3 = whitespace.

RTM export → decompile mapping

Matched: 43 / 43

Export	Decompile line
`ASCIIFILEDLGHOOKPROC`	86850
`ASCIIFORMATDLGPROC`	86994
`ASYM_ASCIIFILEDLGHOOK`	86812
`ASYM_CDROMSAHPCBK`	78334
`BOOKQUERYSAVE`	22225
`CACHEFILEOPTIONSDIALOGPROC`	78444
`CDROMSAVEASHOOK`	78356
`CHANGEPRINTERDIALOGPROC`	136971
`CONDITIONDIALOGPROC`	132432
`DDEWNDPROC`	28773
`EGCDBSAVEBOOK`	27357
`EGDYNALINKERRORFUNC`	40852
`EGYPTCHILDWNDPROC`	47953
`EGYPTCLIENTWNDPROC`	31331
`EGYPTNOTIFYPROC`	32680
`ENUMWINSFOROPEN`	46390
`FONTDLGHOOKPROC`	30737
`FONTSTATUSWNDPROC`	138995
`HANDLEREMOVESCRIPTPROC`	72783
`HEADERDIALOGPROC`	137807
`HEADERNOTIFYPROC`	138403
`HEADFOOTWNDPROC`	66993
`IMAGESTORESTATUS`	28163
`MAINWNDPROC`	37122
`MMRESDLGCHOOSE`	150899
`OPTIMIZEDSAVESTATUSDIALOGPROC`	78670

Export	Decompile line
`PARAGRAPHDIALOGPROC`	83266
`PLASKDIALOGPROC`	34880
`PLREQUESTDIALOGPROC`	34410
`POSTPALCHNGTOOTHERS`	73918
`PREVIEWNOTIFYPROC`	138525
`PRINTOPTIONSDIALOGPROC`	135102
`PRINTPAGEDIALOGPROC`	134383
`PRINTREPORTDIALOGPROC`	133543
`RESCHOOSEDLGPROC`	148133
`RESOURCELISTENUM`	93087
`SEARCHDIALOGPROC`	84108
`STATUSBARWNDPROC`	54892
`SUBEDITCONTROLPROC`	55062
`TOOLNOTIFY`	59326
`XCTOLEREMOTECOMMAND`	103260
`XCTOLEREMOTEGET`	103345
`XDLLTRANSWINPROC`	88606

W16 export → decompile mapping

Matched: 71 / 72

Export	Decompile line
`ACXCTBKSIMPLEOLEDISPATCH_DELETE`	8697
`ACXCTBKSIMPLEOLEDISPATCH_NEW`	8667
`ACXFREEDATAOBJECT`	8583
`ACXGETDATAOBJECT`	8519
`ACXGETTOOLBOOKSUPPORTCONFIGURATION`	8202
`ACXINIT`	8446
`ACXINITIALIZETOOLBOOKSUPPORT`	8338
`ACXOLEINITIALIZE`	8612
`ACXOLEUNINITIALIZE`	8642
`ACXSTARTUP`	8427
`ACXUNINITIALIZETOOLBOOKSUPPORT`	8376
`ASYMDIBUTILCONVERT16`	22541
`ASYMGETVDMPOINTER16`	10205
`ASYMRELEASEVDMPOINTER16`	10236
`ASYM_CGETCWD`	18704
`ASYM_CLSIDFROMPROGID`	16879
`ASYM_CLSIDFROMSTRING`	16929
`ASYM_CONTEXTOPENFILE`	19821
`ASYM_CONVERTTOTREATAS`	17105
`ASYM_CREATEBITMAPFROMOLEICON`	16564
`ASYM_DETERMINEGUIDTYPE`	17198
`ASYM_DLLREGISTERSERVER`	16497
`ASYM_ENUMICONDRAW`	16804

Export	Decompile line
`ASYM_FREEOLEICON`	16838
`ASYM_GETOPENFILENAME`	18945
`ASYM_GETSAVEFILENAME`	19315
`ASYM_GETSAVEFILENAMEEXPLORER`	19318
`ASYM_IS32BITENV`	18663
`ASYM_OLEICONDRAW`	16622
`ASYM_OPENFILE`	19652
`ASYM_OPENFILEEX`	19903
`ASYM_STRINGFROMCLSID`	16867
`ASYM_URLDOWNLOAD`	19972
`ASYM_XFORMHEIGHTINHIMETRICTOPIXELS`	16724
`ASYM_XFORMHEIGHTINPIXELSTOHIMETRIC`	16700
`ASYM_XFORMWIDTHINHIMETRICTOPIXELS`	16678
`ASYM_XFORMWIDTHINPIXELSTOHIMETRIC`	16654
`ASYM__CHDIR`	18747
`ASYM__FULLPATH`	18803
`ASYM__GETCWD`	18730
`ASYM__LCREAT`	19631
`ASYM__LOPEN`	19688
`ASYM__MAKEPATH`	19727
`CALLPROC32W`	4136
`COMPILERCOMPILETEXT`	12395
`COPYDITHERED8BITPALETTE32`	22264
`CREATEDIBMASK32`	22242
`FINDORLAUNCHREMOTECALLSERVER`	8115
`FREELIBRARY32W`	4074
`GETMODULEFILENAME32W`	6913
`GETPLATFORMREF`	11072
`GETPROCADDRESS32W`	4094
`GETRUNTIMELOCALIZEDDLL`	10638
`GETTHUNKINGLAYERSERRORFUNC`	5226
`GETVDMPOINTER32W`	4114
`IS32BITENV`	3892
`ISCOMMONDLGLAUNCHED`	18475
`ISWINDOWS95ORLATER`	9922
`LOADLIBRARYEX32W`	4053
`LOADLOCALIZEDSTRING`	5032
`MINIMIZETHUNKINGLAYERS`	5133
`OPENFILE32W`	6943
`REGQUERYINFOKEY32W`	11759
`REGQUERYVALUEEX32W`	11713
`REGSETVALUEEX32W`	11679
`SCRIPTED`	13053
`SETTHUNKINGLAYERSERRORFUNC`	5217
`SHUTDOWNTHUNKINGLAYERS`	3919
`SYSTEMPARAMETERSINFO32W`	11355
`WINHELP32W`	11260

Export	Decompile line
`WOWHANDLE32`	4418

Unmatched

• `WEP`

BAS ordinal map (from ExportList.java)

Source: Ghidra ExportList.java console output.

Ordinal	Name	Address
1	`CDBINIT`	`1058:0000`
2	`CDBCREATEINSTANCE`	`1038:0000`
3	`CDBDESTROYINSTANCE`	`1038:040a`
4	`CDBCREATETHREAD`	`1038:0320`
5	`CDBDESTROYTHREAD`	`1038:047a`
6	`CDBCLIENTSTATUS`	`1038:04c6`
7	`CDBFILENEW`	`1048:0028`
8	`CDBFILEOPEN`	`1048:017a`
9	`CDBFILESAVE`	`1048:030a`
10	`CDBFILESAVEAS`	`1048:04ae`
11	`CDBFILECLOSE`	`1048:0948`
12	`CDBFILEPATH`	`1048:09e0`
13	`CDBFILEMERGE`	`1048:0a18`
14	`CDBVIEWBOOK`	`1050:00f0`
15	`CDBVIEWPAGE`	`1040:0000`
16	`CDBEDITBOOK`	`1050:007e`
17	`CDBVIEWPAGEVALUE`	`1040:041e`
18	`CDBVIEWBKGND`	`1040:01f2`
19	`CDBGETSELECTPROP`	`1068:0000`
20	`CDBCLONEINSTANCE`	`1038:009a`
21	`CDBTHREADFROMINSTANCE`	`1038:036c`
22	`CDBFILESTATUS`	`1050:0000`
23	`CDBVIEWNONE`	`1040:02f8`
24	`CDBSENDSELECT`	`1068:009e`
25	`CDBSETSELECTPROP`	`1068:0686`
26	`CDBPAINTCLIENT`	`1010:0506`
27	`CDBDISPLAYSTATE`	`1010:0000`
28	`CDBSENDUNDO`	`1068:07ca`
29	`CDBCREATEBOOKREF`	`1048:0f5c`
30	`CDBCREATEPAGEREF`	`1040:03b2`
31	`CDBCREATEBKGNDREF`	`1040:0666`
32	`CDBCREATEGRAPHICREF`	`1078:0076`
33	`CDBREFVALUE`	`1078:0304`
34	`CDBDEREFVALUE`	`1078:0400`
35	`CDBSAMEVALUEP`	`1078:049e`

Ordinal	Name	Address
36	`CDBPARENTVALUE`	`1078:07b2`
37	`CDBQUERYVALUE`	`1078:0958`
38	`CDBCLEARBOOKREFS`	`1048:10a6`
39	`CDBGETVALUE`	`1078:0d54`
40	`CDBSETVALUE`	`1078:1450`
41	`CDBSENDVALUE`	`1078:1110`
42	`CDBQUERYMULTISAVE`	`1048:0b90`
43	`CDBMULTISAVE`	`1048:0e24`
44	`CDBSENDFOCUS`	`1060:0000`
45	`CDBGETFOCUSPROP`	`1060:019a`
46	`CDBSETFOCUSPROP`	`1060:01d6`
47	`CDBSETFOCUS`	`1060:0244`
48	`CDBBOOKOFVALUE`	`1078:0612`
49	`CDBLAYEROFVALUE`	`1078:06fe`
50	`CDBQUERYFILEPATH`	`1078:0aaa`
51	`CDBSENDCREATE`	`12c8:00bc`
52	`CDBNEWOBJECT`	`12c8:02c2`
53	`CDBCHUNKEND`	`12f0:0000`
54	`CDBCHUNKFIND`	`12f0:0146`
55	`CDBSKIPSEP`	`12f0:04b4`
56	`CDBCHUNKLAST`	`12f0:0334`
69	`CDBCHUNKREMOVE`	`12f0:0550`
228	`OBJLOCK`	`1170:0308`
233	`OBJNEWNEAR`	`1170:0552`
355	`CDBCHECKBOUNDDTL`	`12c8:0000`
405	`TBXEXECSCRIPT`	`1268:060c`

Key entries for bounds work:

• Ordinal 51 `CDBSENDCREATE` at `12c8:00bc`

• Ordinal 52 `CDBNEWOBJECT` at `12c8:02c2`

• Ordinal 355 `CDBCHECKBOUNDDTL` at `12c8:0000`

Full export list (606 ordinals) was provided by the user.

This table holds the confirmed mappings relevant to the current research.